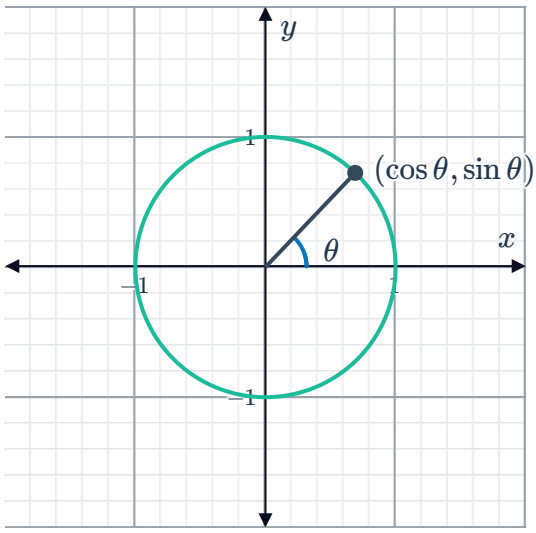
Graphs of trigonometric functions

1 The value of $\sin\theta$ can be represented on the given xy-plane.

a Consider the curve $y = \sin\theta$:
Complete the following:



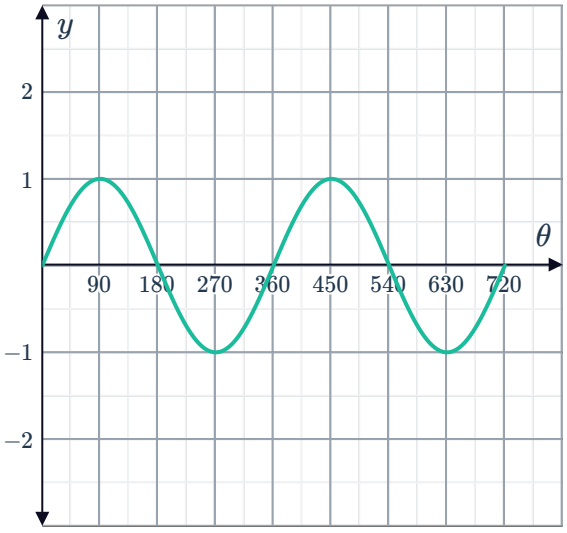
i A rotation of 380° has the same rotation as the acute angle with measure $\square$. So $\sin 380^\circ = \sin\square$.

ii A rotation of 480° has the same rotation as the obtuse angle with measure $\square$. So $\sin 480^\circ = \sin\square$.

iii A rotation of 580° has the same rotation as the reflex angle with measure $\square$. So $\sin 580^\circ = \sin\square^\circ$.

b Consider the graph of $y = \sin\theta$ for $0 \le \theta \le 720^\circ$:
Describe the nature of $y = \sin\theta$.

c The number of degrees it takes for the curve to complete a full cycle is called the period of the function.
Determine the period of $y = \sin\theta$.



2 Consider the curve $y = \sin x$ and state whether the following statements are true or false:

a The graph of $y = \sin x$ is periodic.

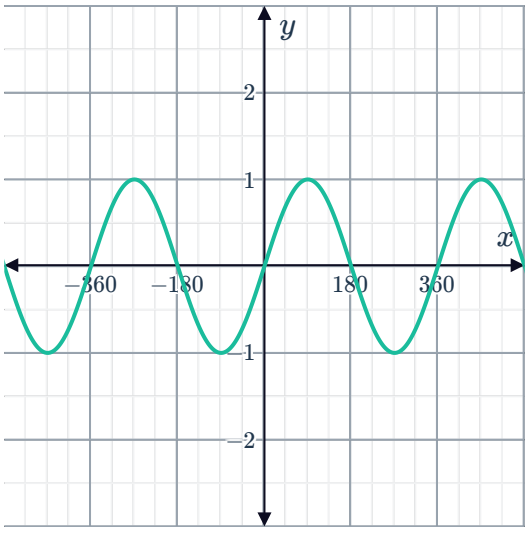
b As x approaches infinity, the height of the graph approaches infinity.

c The graph of $y = \sin x$ is increasing between $x = (-90)^\circ$ and $x = 0^\circ$.

d The graph of $y = \sin x$ is symmetric about the line $x = 0$.

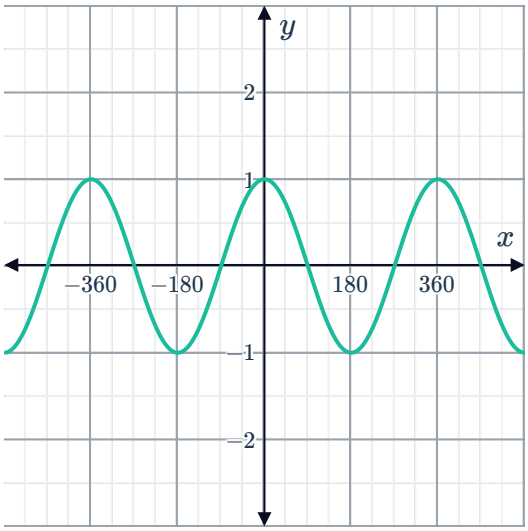
e The graph of $y = \sin x$ is symmetric with respect to the origin.

f The y-values of the graph repeat after a period of 360°.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

3 Consider the curve $y = \cos x$ and state whether the following statements are true or false:



a The graph of $y = \cos x$ is cyclic.

b As x approaches infinity, the height of the graph approaches infinity.

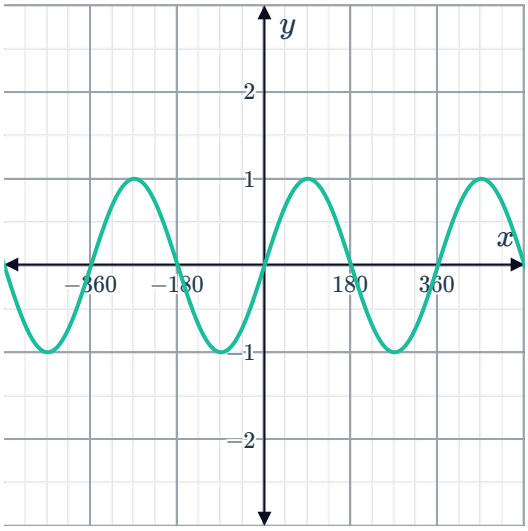
c The graph of $y = \cos x$ is increasing between $x = (-180)^\circ$ and $x = (-90)^\circ$.

d The graph of $y = \cos x$ has reflective symmetry across the line $x = 0$.

e The graph of $y = \cos x$has rotational symmetric with respect to the origin.

f The y-values of the graph repeat after a period of 180°.

4 Consider the curve $y = \sin x$:

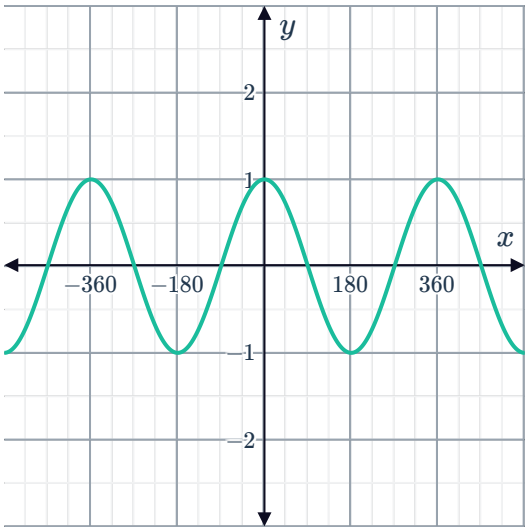


a State the coordinates of the y-intercept.

b Find the maximum y-value.

c Find the minimum y-value.

5 Consider the curve $y = \cos x$:



a State the coordinates of the y-intercept.

b Find the maximum y-value.

c Find the minimum y-value.

6 Consider the equation $y = \cos x$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

a Complete the table of values:

b Sketch the graph of $y = \cos x$.

x	0°	90°	180°	270°	360°
$\cos x$					

7 Consider the equation $y = \sin x$.

a Complete the table of values:

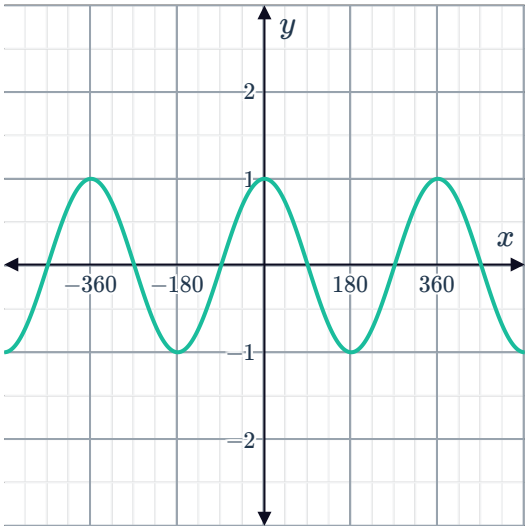
b Sketch the graph of $y = \sin x$.

x	0°	90°	180°	270°	360°
$\sin x$					

8 Consider the graph of $y = \cos x$:

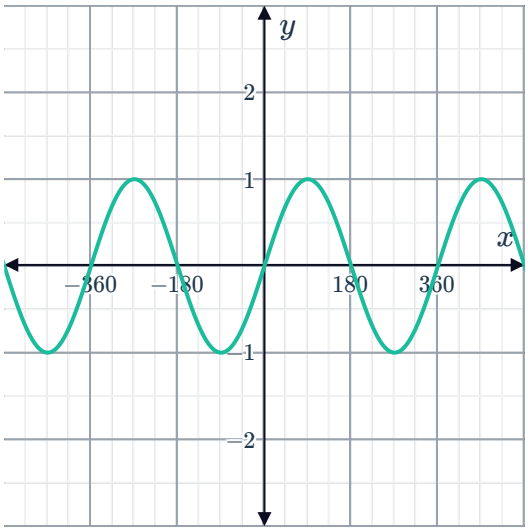
a Is $\cos 162^\circ$ positive or negative?

b In which quadrant does an angle with measure 162° lie in?



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

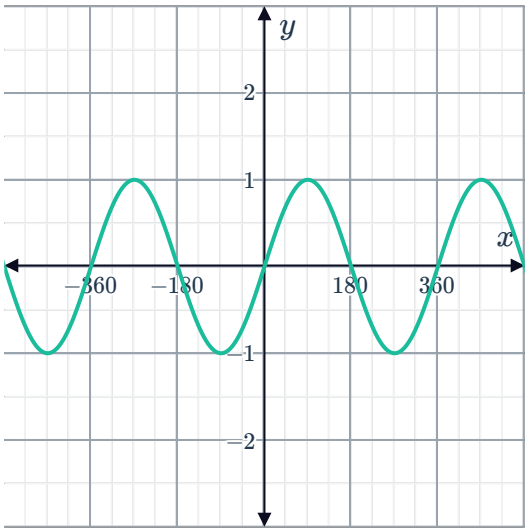
9 Consider the graph of $y = \sin x$:



a Is $\sin 200^\circ$ positive or negative?

b In which quadrant does an angle with measure 200° lie in?

10 Consider the curve $y = \sin x$:



a If one cycle of the graph of $y = \sin x$ starts at $x = 0^\circ$, when does the next cycle start?

b Find the x-value of the x-intercept in the following regions:

i $0^\circ < x < 360^\circ$

ii $(-360)^\circ < x < 0^\circ$

c True or false: As x approaches infinity, the graph of $y = \sin x$ stays between $y = -1$ and $y = 1$.

d State whether the graph of $y = \sin x$ increases or decreases in the following regions:

i $90^\circ < x < 270^\circ$

ii $(-270)^\circ < x < (-90)^\circ$

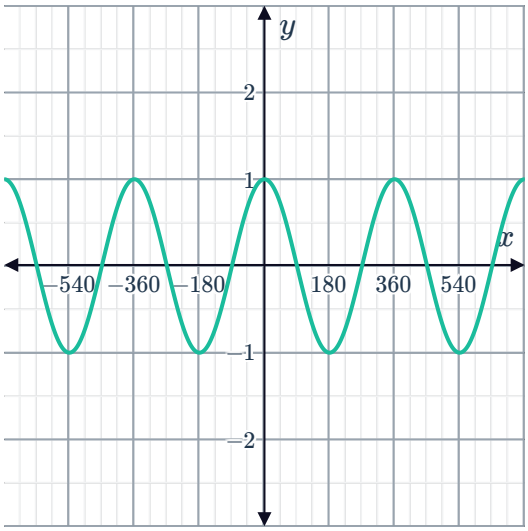
iii $(-450)^\circ < x < (-270)^\circ$

iv $(-90)^\circ < x < 90^\circ$

v $270^\circ < x < 450^\circ$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 Consider the curve $y = \cos x$:



a If one cycle of the graph of $y = \cos x$ starts at $x = (-90)^\circ$, when does the next cycle start?

b Find the x-values of the x-intercepts in the following regions:

i $0^\circ < x < 360^\circ$

ii $(-360)^\circ < x < 0^\circ$

c True or false: As x approaches infinity, the graph of $y = \cos x$ stays between $y = -1$ and $y = 1$.

d State whether the graph of $y = \cos x$ increases or decreases in the following regions:

i $0^\circ < x < 180^\circ$

ii $(-180)^\circ < x < 0^\circ$

iii $180^\circ < x < 360^\circ$

iv $(-360)^\circ < x < (-180)^\circ$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647